

Sentiments

TM the Great Unread

Kristoffer L. Nielbo

DTL|Digital Arts Initiative
Interacting Minds Centre|Aarhus University

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Sentiment analysis

Popular methods for rating the **affective content** of texts

Used in business analytics and bio-NLP to predict market behavior, consumer preferences, happiness and quality of life

Originate in psychometric and sociometric scale studies

Three general approaches:

- ▶ **Dictionary-based methods** (word counting)
- ▶ Supervised learning (machine learning)
- ▶ Unsupervised learning (machine learning)

Dictionary-based methods

A dictionary is basically a set of words with ratings

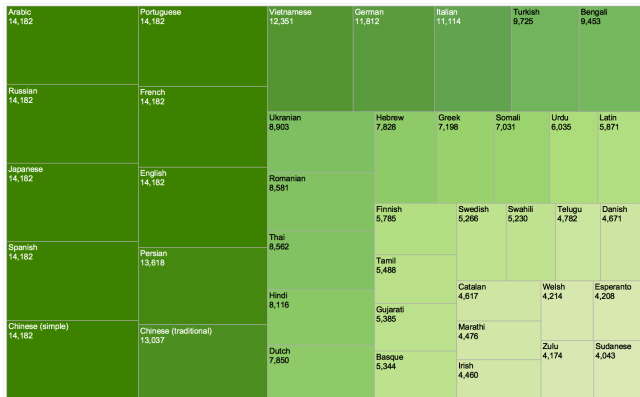
Ratings can be binary (± 1 or 0/1) or based on continuum (1, 2 ... m or 1 : m)

Compute corpus frequency for each dictionary word and multiply their sentiment rating (weight)

Dictionary	# Fixed	# Stems	Total	Range	# Pos	# Neg	Construction	License
LabMT	10222	0	10222	1.3 $\rightarrow$ 8.5	7152	2977	Survey: MT, 50 ratings	CC.
ANEW	1030	0	1030	1.25 $\rightarrow$ 8.82	580	449	Survey: FSU Psych 101	Free for research.
WK	13915	0	13915	1.26 $\rightarrow$ 8.53	7761	5945	Survey: MT, >14 ratings	CC.
MPQA	5587	1605	7192	-1,0,1	2393	4342	Manual + ML	GNU GPL.
LIWC	722	644	1366	-1,0,1	406	500	Manual	Paid, commercial.
Liu	6782	0	6782	-1,1	2003	4779	Dictionary propagation	Free.
PANAS-X	60	0	60	-1,1	10	10	Manual	Copyrighted paper
Pattern 2.6	1528	0	1528	-1,0,+1	528	620	Unspecified	BSD
SentiWordNet 2.6	147701	0	147701	-1 $\rightarrow$ 1	17677	20410	Synset synonyms	CC BY-SA 3.0
AFINN	2477	0	2477	-5, -4, ..., 4, 5	878	1598	Manual	ODbL v1.0
General Inquirer	4205	0	4205	-1,+1	1915	2290	Harvard-IV-4	Unspecified
WDAL	8743	0	8743	1 $\rightarrow$ 3	6517	1778	Survey: Columbia students	Unspecified
NRC	1220176	0	1220176	-5 $\rightarrow$ 5	575967	644209	PMI with emoticons	Free for research

Languages

Number of entries in the NRC Emotion Lexicon, By Language



Number of words



Pros and cons

Advantages (in comparison to ML)

- ▶ Corpus agnostic (can be applied without training)
- ▶ Avoid *black boxing* the solution

Assumptions and problems

- ▶ Bag-of-words assumption
- ▶ Large data: Accuracy depends on large data set (single sentence or paragraphs are useless)
- ▶ Contextual errors: Context sensitivity of word meaning ($miss_{\downarrow}$, $vice_{\downarrow}$) and negations ($\{not_{\downarrow} good_{\uparrow}\}_{neutral}$)
- ▶ Lower accuracy than supervised learning (but supervised learning needs class information and is corpus dependent)

Word rating

Words in dictionaries are rated by according to more or less principled procedures:

- ▶ Survey-based: Random samples or crowd sourcing (MTurk)
- ▶ Manual: expert or naive ($\sim$ convenience)

Rating issues

- ▶ Space and time specificity (e.g., ANEW is from 2000)
- ▶ Dependencies between raters
- ▶ The *WIERD* problem (LIWC was based on American undergraduates)

Mismatches

Across dictionaries we find words that seem incorrectly rated

Negative_{MPQA} : {moonlight, cutest, finest, funniest, comedy, laugh}*

Positive_{LIWC} : {dynamite, careful, richard, silly, gloria, securities, boldface}*

- ▶ Reliance on specific sample of raters
- ▶ 'Dirty' ratings

Literary applications

